# Paleolithic technology paths

From our ~200,000-year starting point to the end of the last Ice Age, roughly 11,700 years ago. **No Neolithic progression.**

**How to read this report.** "Starting toolkit" means appropriate knowledge, not spawn equipment. Regional traditions and materials matter. Finds establish availability, not exact invention dates.

† **Plausible reconstruction:** a design allowance, not a securely demonstrated practice at the stated date. Regional period names vary; pre-cutoff Epipaleolithic examples are included.

## 01 Weapons

**STARTING TOOLKIT · ~200,000 YEARS AGO**

Handheld stone cutting tools, wooden spears and stone-tipped spear technology. Simple clubs†.

**LATER PALEOLITHIC UNLOCKS**

Bows and arrows, spear-throwers and specialized stone or bone points. Slings and wooden/hide shields† are optional inventions, not an established standard kit. [01], [03]

## 02 Hunting, trapping & fishing

**STARTING TOOLKIT · ~200,000 YEARS AGO**

Hunting spears, digging implements and butchery tools. Simple capture barriers†.

**LATER PALEOLITHIC UNLOCKS**

Barbed fishing points, harpoons and hooks; hooks survive from about 23,000 years ago. Netting is supported, with some uses inferred. Basket traps, weirs, snares and deadfalls† are optional reconstructions. [07], [15]

## 03 Medicine & injury care

**STARTING TOOLKIT · ~200,000 YEARS AGO**

No securely reconstructable standardized medical kit. Supporting an injured person, protecting wounds and immobilizing injuries are plausible practices†.

**LATER PALEOLITHIC UNLOCKS**

Dressings, splints and locally learned plant preparations†. A roughly 14,000-year-old tooth shows cavity scraping; this supports exceptional intervention, not routine dentistry or proven herbal cures. [13]

## 04 Clothing & body protection

**STARTING TOOLKIT · ~200,000 YEARS AGO**

Hide scraping and preparation; simple draped skins or wraps†.

**LATER PALEOLITHIC UNLOCKS**

Awls, eyed needles and fitted sewn hide garments. Footwear, belts, head coverings and hide padding† are sensible design options; exact early forms are often reconstructed. No default loom-made linen industry. [08]

# Crafting, food & materials

These processes enable several kinds of equipment. They should not be hidden inside a single enormous “tools” menu.

## 05 Carrying, containers & storage

**STARTING TOOLKIT · ~200,000 YEARS AGO**

Simple hide bundles and useful natural containers†.

**LATER PALEOLITHIC UNLOCKS**

Hide bags, baskets and bark/wood vessels†; portable forms are often poorly preserved. Simple pottery is a regional late unlock, roughly 20,000–16,000 years ago. Framed carriers and food caches†; no farming granaries. [11], [15]

## 06 Food preparation & preservation

**STARTING TOOLKIT · ~200,000 YEARS AGO**

Butchery, cracking and pounding foods, and cooking over fire.

**LATER PALEOLITHIC UNLOCKS**

Grinding and wild-plant processing; bread-like foods occur about 14,400 years ago. Drying, smoking, hot-stone boiling and earth-oven cooking† are practical branches with uncertain early recipes. No dairying. [09], [12]

## 07 Fire, lighting & heating

**STARTING TOOLKIT · ~200,000 YEARS AGO**

Hearths, maintaining fire and carrying embers†. Deliberate ignition where technique and suitable materials are known.

**LATER PALEOLITHIC UNLOCKS**

Improved hearths, torches and fat lamps. Friction-fire kits† are a reasonable craft allowance, with uncertain method-specific early dates. Distinguish maintaining fire from reliably starting it. [02]

## 08 Tools & material processing

**STARTING TOOLKIT · ~200,000 YEARS AGO**

Hammerstones, flakes, handaxes, scrapers, worked wood and basic bone implements.

**LATER PALEOLITHIC UNLOCKS**

Specialized blades, awls, drills and bone/antler tools. Ground-edge axes occur regionally in Australia by about 44,000–49,000 years ago; this does not imply universal Neolithic-style carpentry. [01], [10], [16]

## 09 Fibre, adhesives & ceramics

**STARTING TOOLKIT · ~200,000 YEARS AGO**

Basic woodworking and material selection; some composite-tool technology exists. Do not assume every group has a sophisticated adhesive recipe.

**LATER PALEOLITHIC UNLOCKS**

Twisted cord, rope, plaiting, twining and netting; fibre impressions survive from the Upper Paleolithic. Better hafting adhesives and regional clay firing expand the toolkit. Pottery need not imply agriculture. [06], [11], [15]

# Camp, movement & wild resources

Mobile hunter-gatherers can have sophisticated crafts and well-organized homes. This scope does not assume the architecture or economy of later farming villages.

## 10 Shelter & construction

**STARTING TOOLKIT · ~200,000 YEARS AGO**

Windbreaks, simple branch shelters and rough shelter frames†; using and improving natural shelters.

**LATER PALEOLITHIC UNLOCKS**

Brushwood huts with hearths and bedding are evidenced. Pole-and-hide tents and earth-covered shelters† fit progression; village architecture is outside scope. [04], [12]

## 11 Furniture & camp fittings

**STARTING TOOLKIT · ~200,000 YEARS AGO**

Grass bedding; conveniently placed logs or stones for sitting†.

**LATER PALEOLITHIC UNLOCKS**

Prepared plant bedding, hides, mats†, simple racks, pegs and sleeping platforms†. Formal chairs and fitted stone beds are not established defaults. [05], [12]

## 12 Travel, hauling & watercraft

**STARTING TOOLKIT · ~200,000 YEARS AGO**

Walking with carried loads; improvised hauling arrangements†.

**LATER PALEOLITHIC UNLOCKS**

Drag skids, carrying frames and simple watercraft†. Ancient sea crossings do not reveal an exact boat design. No livestock traction, dog-sled teams or wheeled vehicles.

## 13 Water & camp sanitation

**STARTING TOOLKIT · ~200,000 YEARS AGO**

Collecting water from available sources. Separating waste from sleeping areas†.

**LATER PALEOLITHIC UNLOCKS**

Water containers†, settling/straining† and hot-stone boiling†. Camp waste areas are evidenced. No lined village wells; straining alone does not disinfect water. [12]

## 14 Wild resources & animal relationships

**STARTING TOOLKIT · ~200,000 YEARS AGO**

Wild-food gathering, digging and knowledge of local plants and animal behaviour.

**LATER PALEOLITHIC UNLOCKS**

Wild-seed processing, seasonal harvesting and caches†. Dogs are present by about 14,200 years ago. Farming, livestock herds and milking are outside scope. [12], [14]

# What can we actually build?

Build around shelters, hearths and camp fittings. Complexity can grow substantially without importing Neolithic houses. Entries marked † are deliberate gameplay reconstructions.

| BUILDING | RECOMMENDATION |
|---|---|
| Branch-and-leaf lean-to | Starting option†. Modest protection and limited durability. |
| Conical pole-and-hide tent | Include after adequate hide preparation and bindings. Use a generic description rather than claiming the specific historical tipi design for every prehistoric group. |
| Brushwood hut | Clear inclusion. Ohalo II preserves roughly 23,000-year-old huts, hearths, working areas and bedding. [12] |
| Reed-roofed pole shelter | Reasonable regional design† with suitable poles, bindings and roofing plants. Exact form and weather resistance need testing. |
| Earth-covered shelter | Optional design†. Drainage, roof loads and available timber limit practicality; not a universal prehistoric house type. |
| Bone-supported shelter | Possible late regional branch†. Do not interpret every mammoth-bone ring as a complete dwelling; function can be disputed. |
| Stone windbreak / low wall | Modest camp improvement†. This is not evidence for importing later stone-built village houses or fitted stone furniture. |
| Timber cabin / longhouse | Exclude as a standard unlock. Earlier worked or joined wood proves woodworking capability, not a particular log cabin or Neolithic house plan. [4] |
| Snow shelter / classic igloo | Simple snow windbreaks or dug shelters† may be design allowances. Do not call the much later documented block-built Inuit igloo an established Paleolithic technology. |

**Beds are a clear inclusion.** Prepared grass bedding is evidenced around 200,000 years ago. Upgrade its warmth, dryness and parasite control before adding framed furniture†. A bed does not require a modern bed frame. [5]

**Medicine needs restraint.** Care and wound protection† make useful mechanics. Exact ancient treatments and their effectiveness are rarely recoverable; isolated dental intervention does not justify generic healing potions. [13]

# Design the unlocks as processes

Keep three constraints separate: **technology** is knowing the process; **ability** is performing it well; **materials** determine what is possible locally. Culture affects transmission and established practice.

**Cordage:** Stronger bindings, carrying straps, net-making and shelter frames.

**Hide preparation & sewing:** Fitted clothing, bags, coverings and bedding.

**Woodworking & hafting:** Better tools, spears and shelter components; sophistication need not mean cabins.

**Grinding & controlled heating:** Wild-food processing and cooking; regional pottery is a late branch, not a farming prerequisite.

Knowing how to make a bow is technology; making a dependable bow and shooting accurately are different skills. Recipes should use properties such as suitable fibres, workable timber and knappable stone—not require particular real-world species names.

**Recommended scope:** start with stone/wood tools, spears, fire handling, hide preparation and plant bedding. Add cordage, sewn clothing, specialized hunting/fishing and better shelters. Reserve pottery and dogs for regional late progression; omit farming, dairying, village infrastructure, wheels and metalworking.

# Evidence & further reading

[01] Smithsonian — Stone tools

[02] Nature — Fire-making evidence around 400,000 years ago

[03] PLOS ONE — Early stone-tipped projectiles

[04] Nature — Structural wood use at Kalambo Falls

[05] Science / SUNY — Grass bedding around 200,000 years ago

[06] Scientific Reports — Neanderthal fibre technology

[07] PNAS — Paleolithic shell fishhooks, Okinawa

[08] Smithsonian — Making clothing

[09] PNAS — Bread-like foods before agriculture, 14,400 years ago

[10] Australian Archaeology — Ground-edge axes, 44,000–49,000 years ago

[11] Science — Early pottery in China; earliest dating remains debated

[12] Journal of Lithic Studies — Ohalo II huts, tools and food processing

[13] Scientific Reports — Upper Paleolithic dental intervention

[14] Journal of Archaeological Science — Bonn-Oberkassel dogs

[15] PLOS ONE — Upper Paleolithic fishing and netting evidence

[16] Nature — Systematic bone tools at 1.5 million years

Evidence establishes particular populations and practices, not universal knowledge. † entries are explicit design inferences; missing organic remains do not demonstrate that a technology was absent.